# TECHNOLOGY STACK

**Project Name:** Implement Client Script & UL Policy
**Project ID:** SWTID-2026-3268

## Proposed Technology Architecture

| Layer | Technology / Component | Purpose |
|---|---|---|
| Client Layer | JavaScript / TypeScript | Implement client-side workflow, validation, and policy interaction. |
| UI Layer | HTML5 / CSS3 / Existing UI Framework | Present controls, status, messages, and user interactions. |
| Policy Layer | UL Policy Rules / Configuration | Define and enforce the required policy conditions. |
| Validation Layer | Client-side validation + policy checks | Validate inputs and determine whether requests meet policy requirements. |
| API / Integration | REST API / HTTPS | Exchange required data with approved backend or policy services. |
| Backend | Existing approved application services | Process server-side operations and integrations where required. |
| Data / Audit | Approved database or logging service | Store required configuration, audit events, and troubleshooting information. |
| Testing | Unit, integration, and end-to-end testing | Verify functionality, policy compliance, compatibility, and regression behavior. |
| Version Control | Git-based repository | Manage source code, configuration changes, reviews, and release history. |
| Deployment | Approved CI/CD or release process | Build, validate, and deploy the client script and related configuration. |

## Technology Selection Principles

• Prefer technologies already approved and supported by the project environment.
• Keep client-side policy logic modular and maintainable.
• Use secure HTTPS communication and appropriate access controls.
• Minimize unnecessary client-side data exposure.
• Maintain version control, automated testing, logging, and documented deployment practices.

## Architecture Flow

**Client UI** → **Client Script** → **Validation & UL Policy** → **API / Backend Services** → **Data / Audit Logging**